

PRD_Documents_Module

Documents Module and Document Page

Full Product Requirements Document for Product, UX, Frontend, Backend, QA, Security and DevOps Teams

Design principle: combine a predictable folder system with powerful global document discovery, while every document remains one controlled record with a permanent Document Passport, linked business context and complete revision history.

Status: Confirmed current-scope specification

Version: 1.0

Third-party integrations: Deferred to future roadmap

1. Document Control

Field	Value
Document	PRD_Documents_Module - Documents Module and Document Page
Audience	Product, UX/UI, Frontend, Backend, QA, DevOps, Security and Data teams
Primary pages	Documents Module Page and Document Page
Core supporting flows	Universal Upload, Revision Upload, Review, Approval Request, Promote from Tranzit, Move to Official Folder, Archive, Restore, Compare Revisions, Create Shortcut
Current access approach	All authenticated user types are supported. A detailed role and permission matrix will be applied later without redesigning the module.
Integration policy	No third-party integrations in the initial implementation. Architecture remains integration-ready.

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3. Product Vision and Outcomes

The Documents Module is the platform-wide source of truth for files, drawings, contracts, reports, forms, images, models and records. It is not a detached file drive. Each document is connected to business entities and workflows across Projects, Units, Tasks, Meetings, Approvals, Clients, Finance, Government & Legal, HR and other configured modules.

- Enable every user type to find the correct document quickly, whether they browse folders or search globally.
- Maintain one underlying document record even when the document appears in multiple folders or business contexts.
- Preserve every revision and clearly distinguish draft, review, approved, issued and superseded states.
- Provide a collaborative working area through Project Tranzit without contaminating official folders.
- Support a default platform structure while allowing custom folders and module-specific systems such as HR.
- Create a self-contained first release with no dependency on external cloud, signature, OCR or AI services.
- Keep the architecture ready for future Microsoft 365, Google Workspace, digital signature, OCR, AI and BIM integrations.

The module should feel like a premium business knowledge system, not a generic file manager.

4. Scope and Non-Goals

4.1 In scope

- Hybrid folder and global filtered views
- System and custom folders
- Project Tranzit folders
- Personal Workspace
- Document Passport
- Document Page
- Revision history
- Comments and mentions
- Review and approval linking
- Internal file preview
- Tags and metadata
- Favorites and recent documents
- Document collections
- Templates
- Expiration tracking
- Required reading
- Watermarks
- Bulk operations
- Archive and restore
- Duplicate detection
- Broken-link and orphan repair
- Storage dashboard
- Responsive layouts and premium animations

4.2 Deferred

- Microsoft 365, Google Workspace or external storage integrations
- DocuSign, Adobe Sign or qualified electronic signature integrations
- External OCR service integration
- AI classification and semantic search
- External BIM/CDE integrations

- Email ingestion and external mail delivery
- Public external share links with anonymous access

5. User Types and Use Cases

This PRD includes every platform user category. The first implementation exposes the feature set broadly to authenticated users. Detailed visibility, edit, delete, approval and download restrictions will be introduced later through the permission system.

User type	Primary document needs
Owner / CEO / Group Leadership	Cross-company visibility, executive records, contracts, approvals, reports and audit history
Company Admin / Operations	Folder governance, storage overview, classification, bulk operations and lifecycle management
Project Manager	Project folders, drawings, reports, meeting documents, Tranzit, approvals and issue tracking
Architect / Designer / BIM	Technical drawings, revisions, comparison, dependencies, comments and approvals
Engineer / QA/QC / HSE	Inspection records, technical reports, site evidence, required reading and issued documents
Finance	Invoices, payment records, budgets, contracts and finance-specific folders
Legal / Government Affairs	Permits, notices, submissions, legal correspondence, expiration dates and controlled issue state
HR	Employee records and HR-specific folder system, required reading and policy documents
Sales / Client Relations	Brochures, offers, floor plans, unit packages, contracts and client-linked documents
Procurement / Warehouse	Purchase documents, specifications, quotations, deliveries and supplier records
Employee / Contributor	Personal Workspace, task-linked documents, project documents and shared review workflows
External participant later	Controlled access to specific records only; implementation deferred to permissions roadmap

6. Information Architecture

- Documents Module Page: global folder tree, filtered document views and document list/grid.
- Document Page: full record for one document, including preview, Passport, revisions, comments, approvals and activity.
- Folder Page / Folder View: system or custom folder context.
- Project Tranzit Folder: collaborative draft workspace.
- Personal Workspace: private staging area for files not yet assigned to a business record.
- Upload Document Flow: upload directly into a selected folder or record.
- Universal Upload Flow: upload from anywhere and choose business context.
- Revision Upload Flow: append a new revision to an existing document.
- Promote from Tranzit Flow: move a working document into an official folder while preserving identity and history.
- Create Shortcut Flow: add a document reference to additional folders without duplicating the file.
- Compare Revisions View: visual/textual comparison between two revisions.

7. Documents Module Page

The Documents Module Page uses a three-zone desktop layout: platform navigation, folder tree and document workspace, with an optional right-side preview/Passport panel.

7.1 Header and global actions

- Page title and description
- Global platform search field
- Upload Document button
- Create Folder action
- View switcher
- Saved View action
- Notification and user controls from the global shell

7.2 Status summary cards

- All Documents
- Recent
- Starred
- Awaiting Approval
- Approved
- Needs Revision
- Expiring Soon
- Archived

7.3 Main views

- List view - default for operational work
- Grid view - visual browsing
- Timeline view - document activity or modification chronology
- Folder view - current selected folder with breadcrumbs

7.4 Table fields

- Selection checkbox
- File icon/thumbnail
- Document title
- Personal Star
- Document code
- Current revision
- Type
- Status
- Project or primary business context
- Last modified date/time
- Quick actions menu

DOC-MOD-001 - Support folder browsing and global filtered views in the same page.

DOC-MOD-002 - Maintain user scope, filters and view when opening and closing the right-side preview panel.

DOC-MOD-003 - Allow a user to Star documents privately.

DOC-MOD-004 - Support saved personal views including folder context, filters, sort, columns and layout.

DOC-MOD-005 - Display a clear empty state with Upload Document and Create Folder actions.

8. Hybrid Folder Architecture

Folders are an organization layer. Business relationships remain stored as document links. A document can be referenced by multiple folders without duplicating the underlying file.

8.1 System folders

- Automatically created from platform records
- Cannot be freely detached from their source record
- May contain standard subfolder templates
- May be renamed only where configuration permits
- Deleting a source record must trigger controlled archive/orphan handling

8.2 Custom folders

- Created by users inside permitted areas
- Can be nested
- Can be renamed, moved and archived
- Do not replace linked business records
- May contain shortcuts to documents stored elsewhere

One document record can appear in multiple folders through references or shortcuts. No file duplication occurs.

9. Default Platform Folder Structure

Root folder	Purpose / example substructure
Company	Corporate governance, policies, licenses, brand, management and company-wide records
Projects	One system folder per project, with configurable project subfolders and mandatory Transit
Buildings / Structures	Optional structure-specific folders under a project
Units	Unit-linked documents, floor plans, contracts, handover and client changes
Clients	Client profiles, offers, correspondence, contracts and shared packages
Tasks	Attachments, deliverables and linked documents for tasks
Meetings	Agenda, minutes, decisions, attachments and action documents
Approvals	Approval packages, supporting documents and approved revision references
Finance	Invoices, payments, budgets, payroll outputs, contracts and finance records
Government & Legal	Permits, licenses, submissions, official correspondence, notices and expiration records
HR	Independent HR folder system for employees, contracts, policies, training and required reading
Personal Workspace	Private staging area owned by the current user

9.1 Project default subfolders

- 01. Project Management
- 02. Drawings
- Architectural
- Structural
- MEP
- Landscape

- 03. Specifications
- 04. Contracts
- 05. Reports
- 06. Site Photos
- Transit
- Archive

Project folder templates must be configurable so other industries can use alternative structures without changing the core document engine.

10. Project Transit Folder

Every Project automatically includes a system folder named Transit. It is a collaborative temporary workspace for drafts, non-final drawings, working files and material under discussion.

- Documents uploaded to Transit default to Transit or Draft status.
- Transit documents are not considered final, approved or officially issued.
- Users can upload, comment, mention participants, create tasks and request review.
- Revisions can be added while the file remains in Transit.
- A document can be promoted to an official folder when ready.
- Promotion preserves the Document ID, revisions, links, comments, approvals and activity history.
- A shortcut may remain in Transit for traceability after promotion, depending on configuration.

TRZ-001 - Create Transit automatically for every Project.

TRZ-002 - Block direct Approved or Issued state on first upload to Transit unless an explicit controlled workflow is used.

TRZ-003 - Provide Promote to Official Folder as a primary action.

TRZ-004 - Record promotion in the document activity timeline.

11. Personal Workspace

Each user receives a private staging area for drafts, screenshots, temporary files and documents not yet assigned to a project or business record.

- Private by default
- Supports folders and tags
- Supports preview and renaming
- Move to Project, Client, Unit, Finance, HR or another official location
- Preserves the document record when moved
- May be used as the starting point for Universal Upload

12. Document Passport

Every document has a permanent Document Passport. The Passport is the metadata and business-context identity of the document, independent of the file bytes or current folder.

Passport group	Fields
Identity	Document ID, title, document code, type, category, file type, primary owner
Organization	Company, project, structure, unit, client, department, source module
Location	Primary folder, additional folder shortcuts, current path
Revision	Current revision, revision status, revision date, previous revision relation
Lifecycle	Draft/Transit/Review/Approved/Issued/Superseded/Archived
Upload	Uploader snapshot, upload timestamp, file size, checksum,

	original filename
Links	Tasks, approvals, meetings, contracts, client changes, finance and legal records
Controls	Confidentiality, required reading, expiration, retention, watermark state
Audit	Created, modified, last opened, activity count and immutable identifiers

A Document Passport persists even when a file is moved, referenced from another folder, revised, archived or promoted from Tranzit.

13. Document Page

The Document Page is the complete workspace for one document.

- Header: title, code, personal Star, revision, status, folder path and primary actions.
- Preview tab: current revision preview and download controls.
- Passport tab: metadata and linked records.
- Revisions tab: full revision list and comparison actions.
- Comments tab: discussion, mentions and review comments.
- Approvals tab: linked approval requests and approval ticket.
- Activity tab: complete chronological document timeline.
- Related documents section: dependencies, collections and related records.
- Quick actions: download, link, create shortcut, upload revision, request review, request approval, promote, archive.

DOC-PAGE-001 - Open the Document Page from any folder, task, approval, meeting or linked record.

DOC-PAGE-002 - Show the exact current revision and lifecycle status at all times.

DOC-PAGE-003 - Allow navigation to source and linked business records.

DOC-PAGE-004 - Use responsive tabs and collapsible information sections on mobile.

14. Document Lifecycle

Status	Meaning
Draft	Initial working state; not ready for formal review
Tranzit	Collaborative project draft under discussion
Internal Review	Being reviewed by selected users
Awaiting Approval	Formal approval request has been submitted
Approved	Specific revision approved but not necessarily issued externally
Issued	Officially released for use or distribution
Needs Revision	Review or approval identified changes
Rejected	Formal approval rejected
Superseded	Replaced by a newer issued or approved revision
Archived	Inactive but retained
Void	Invalidated and not to be used

Status transitions must be configurable but validated. A new revision must not automatically inherit Approved or Issued status from the previous revision.

15. Revisions and Version Control

- A new revision never overwrites the previous revision.
- Each revision has a revision code or number, upload timestamp and uploader snapshot.

- Change summary is mandatory.
- Reason for change and affected sections are supported.
- Approval is attached to the exact revision.
- The latest approved/issued revision is visually distinguished.
- Downloading from a document shortcut always resolves to the current permitted revision unless a specific revision is selected.
- Revision files have checksums for duplicate and integrity checks.

Revision field	Requirement
Revision number/code	Unique within one Document record
File	Immutable uploaded file bytes
Change summary	Required
Reason	Optional/configurable required
Affected sections	Optional structured tags
Status	Revision-specific
Approval	Revision-specific
Previous revision	Direct predecessor relation
Checksum	Required

16. Document References and Shortcuts

- Create shortcut in another folder
- Remove shortcut without deleting document
- Set primary folder
- Show every folder reference in Passport
- Prevent circular folder structures
- Maintain one file and one revision chain
- Allow linked business records independently of folder shortcuts

17. Upload and Universal Upload Flow

17.1 Direct upload

1. Open a folder or source record.
2. Select Upload Document.
3. Choose one or multiple files.
4. Enter title, type, category, code and optional tags.
5. Confirm folder and linked records.
6. Choose Draft or Transit lifecycle state where applicable.
7. Upload and create Document Passport.

17.2 Universal Upload

8. Open Upload Document from any platform page.
9. Select Company and optional Project.
10. Select source module or linked record.
11. Select system folder, custom folder, Transit or Personal Workspace.
12. Complete metadata and upload.
13. System creates folder reference and linked-record relations.

17.3 Upload UX

- Drag and drop
- Multi-file upload

- Upload progress
- Pause/cancel where technically feasible
- Retry failed items
- Duplicate warning
- Metadata defaults from current page context
- Upload summary with success/error rows

18. Promote from Tranzit Flow

14. Open a Tranzit document.
15. Select Promote to Official Folder.
16. Choose destination system/custom folder.
17. Confirm title, type, category, code and current revision.
18. Optionally request approval before or during promotion.
19. System changes primary folder and lifecycle state as configured.
20. Document identity, revision chain, links, comments and activity remain unchanged.

Promotion is a controlled move of the same document record, not a copy operation.

19. Search, Filters and Saved Views

19.1 Search targets

- Title
- Document ID
- Document code
- Original filename
- Tags
- Project/company/client/unit names
- Uploader
- Revision number
- Folder path
- Linked record identifiers
- Extracted text later when OCR/AI is introduced

19.2 Filters

- Company
- Project
- Structure
- Unit
- Client
- Folder
- Document type
- Category
- Status
- Uploader
- Approver
- Revision
- File type
- Date range
- Expiration
- Has comments
- Has approval

- Starred by me
- Archived

19.3 Saved views

- Name
- Owner
- Folder context
- Filters
- Sort
- Layout
- Visible columns
- Optional shared views later

20. Metadata, Tags and Classification

- Document type and category are structured fields.
- Tags are flexible and user-defined.
- Required metadata can vary by source module and folder template.
- Document code format can be generated or manually entered.
- Tags should support autocomplete and normalization.
- The architecture should permit future AI suggestions without making AI required.

21. Comments, Mentions and Review

- Document-level comments
- Revision-specific comments
- Threaded replies
- @user mentions
- Attachments in comments
- Resolve/reopen comment
- Pin formal decision
- Convert comment to Task
- Request review from selected users
- Review completion state
- Comment count and unread state

Mentioning a user should not silently bypass future permissions. The architecture must support access checks when the permission matrix is introduced.

22. Approval Integration

- Request approval from Document Page
- Select exact revision
- Select approver or approval chain
- Add decision request and deadline
- Attach supporting records
- Display approval status in Passport and document list
- Open linked Approval Page
- Add Blue Approval Ticket to the exact approved revision
- Do not transfer approval to a new revision

Approval belongs to the exact revision. A new revision returns to a non-approved state.

23. File Preview and Rendering

File type	Initial preview expectation
PDF	Full-page preview with thumbnails, zoom and page navigation
Images	Zoom, rotate and metadata
Word / Excel / PowerPoint	Server-generated or browser-compatible preview without external service
Video	Basic HTML5 playback where supported
Text / CSV	Readable text/table preview
DWG / DXF	Thumbnail and download initially; advanced native preview may be phased
IFC / GLTF / 3D	Thumbnail and metadata initially; interactive viewer may be phased
Unsupported	File icon, metadata and download

Preview generation should be asynchronous and should not block the upload transaction. Users must still see the document record while preview processing is pending.

24. Revision Comparison

- Select any two revisions
- Side-by-side mode
- Overlay mode for images/PDF pages
- Text difference for supported text documents
- Added/removed/changed highlights
- Page synchronization
- Difference summary
- Export comparison report later

DWG and IFC semantic comparison may be implemented in a later phase, but the data model and UI entry point should be preserved.

25. Document Dependencies and Impact Warnings

- Declare depends-on and affected-by relationships
- Show dependency graph
- Warn when an upstream approved document receives a new revision
- Create review tasks for affected documents
- Link architectural, structural, MEP and other coordinated documents
- Record dependency updates in activity

Example: revising an architectural plan can warn that structural, electrical, plumbing and HVAC documents may require review.

26. Templates and Collections

26.1 Templates

- Contract
- Meeting Minutes
- Site Report
- Inspection Form
- Client Offer
- Material Request

- Purchase Order
- Policy / Procedure
- Custom company templates

26.2 Collections

- Create named group of document references
- One document may belong to multiple collections
- Examples: Sales Package, Handover Package, Permit Submission, Tender Package
- Open/download collection as a group
- Collection does not duplicate files

27. Expiration, Required Reading and Read Receipts

- Expiration date and reminder schedule
- Expiring Soon summary view
- Expired status warning
- Required reading assignment to selected users
- Read/Not Read/Acknowledged status
- Reader timestamp
- Document open receipt
- Completion dashboard for required reading

28. Watermarks and Issue State

- DRAFT
- TRANZIT
- UNDER REVIEW
- APPROVED
- ISSUED
- CONFIDENTIAL
- VOID

Watermarking may be applied dynamically to previews/downloads or generated as controlled derivative files. The original uploaded file should remain unchanged.

29. Bulk Operations

- Move to folder
- Create shortcuts
- Add/remove tags
- Change category
- Request review
- Request approval
- Archive
- Restore
- Download selected
- Add to collection
- Set expiration
- Export metadata

Bulk actions must show a confirmation summary and individual success/failure results.

30. Duplicate Detection and Orphan Repair

- Checksum-based exact duplicate detection
- Filename and metadata similarity warning
- Options: cancel, keep both, create new revision, create shortcut
- Detect broken folder references
- Detect deleted/merged linked records
- Repair or reassign orphaned links
- Never silently delete the underlying document because a linked business record is removed

31. Archive, Restore and Retention

- Soft archive by default
- Archived view
- Restore to original or selected folder
- Retention category and retention end date
- Legal hold field for future use
- Permanent deletion separated from normal archive
- Archive activity and actor recorded

32. Storage Dashboard

- Total storage
- Storage by company
- Storage by project
- Storage by module
- Largest files
- File type distribution
- Storage growth over time
- Preview/derivative storage
- Archived storage
- Failed preview jobs
- Duplicate savings estimate

33. Notifications

- Document uploaded to linked record
- Mention
- Comment reply
- Review requested
- Approval requested
- Approval result
- New revision
- Document issued
- Document superseded
- Expiration approaching
- Required reading assigned
- Required reading overdue
- Transit document promoted

34. Responsive Design

34.1 Desktop

- Persistent platform sidebar
- Folder tree column
- Main document workspace
- Optional right preview/Passport panel
- Resizable columns where practical

34.2 Tablet

- Collapsible platform sidebar
- Folder tree as drawer
- Preview panel as slide-over
- Responsive summary cards
- Scrollable filter bar

34.3 Mobile

- Single-column list/grid
- Folder browser in bottom sheet or full-screen drawer
- Sticky search and Upload action
- Document details as full-screen page
- Preview, Passport, Revisions, Comments and Activity as tabs
- Bottom action bar for Download, Comment, Upload Revision and More

Core document actions must remain available on mobile; the module must not become read-only.

35. Premium Motion and Micro-interactions

- Folder hover: 120-160 ms warm tint, icon emphasis and subtle horizontal movement.
- Document row hover: 120-180 ms soft lift, border tint and quick-action reveal.
- Summary card hover: 2-4 px lift, shadow increase and restrained icon scale.
- Preview panel: 220-280 ms slide and opacity transition.
- Folder expansion: 160-220 ms height and chevron rotation.
- Star toggle: short scale/fade with gold highlight.
- Upload progress: smooth progress interpolation and success confirmation.
- Status change: chip color transition and short menu reveal.
- Tab switching: underline glide and content crossfade.
- Revision upload: optimistic timeline insertion after server confirmation.
- Drag/drop upload zone: border glow and subtle scale response.
- Move/promote flow: document card transitions to destination breadcrumb confirmation.
- Loading: skeletons matching folder tree, list rows, cards and preview layout.

All motion must respect reduced-motion settings and must never delay core actions.

36. Accessibility

- Keyboard navigation for folder tree and document list
- Visible focus states
- Screen-reader labels for file types, status and actions
- Color is not the only status indicator
- Accessible table headers and sorting state
- Reduced-motion support

- Sufficient contrast
- Preview controls keyboard accessible
- Alt text for document thumbnails where possible

37. Data Model

Entity	Purpose	Key fields
Document	Permanent business record	id, tenant_id, company_id, title, code, type_id, category_id, status, primary_folder_id, owner_id
DocumentRevision	Immutable revision	id, document_id, revision_code, file_object_id, checksum, change_summary, status, uploaded_by, uploaded_at
FileObject	Stored binary and derivatives	id, storage_key, original_name, mime_type, size, checksum, preview_state
Folder	System/custom hierarchy	id, parent_id, folder_type, source_entity_type, source_entity_id, name, archived_at
DocumentFolderRef	Folder shortcut relation	document_id, folder_id, is_primary, created_by, created_at
DocumentLink	Generic business relation	document_id, entity_type, entity_id, relation_type, is_primary
DocumentTag	Tag relation	document_id, tag_id
DocumentComment	Discussion	document_id, revision_id, author_id, body, parent_id, resolved_at
DocumentApprovalLink	Approval relation	document_id, revision_id, approval_id
DocumentDependency	Document graph	document_id, depends_on_document_id, dependency_type
DocumentCollection	Named reference package	id, name, owner_id, company_id
DocumentCollectionItem	Collection membership	collection_id, document_id, revision_id, sort_order
DocumentFavorite	Private Star	document_id, user_id, created_at
DocumentReadReceipt	Open/required reading	document_id, revision_id, user_id, opened_at, acknowledged_at
DocumentActivity	Timeline event	document_id, revision_id, actor_id, event_type, payload_json, created_at
DocumentSavedView	Personal view	user_id, name, folder_id, layout, filters_json, sort_json, columns_json

38. Storage and Backend Architecture

- Object storage abstraction for file bytes
- Relational database for Passport, folders, links and activity
- Preview/thumbnail worker queue
- Checksum service
- Virus/malware scanning hook
- Search indexing pipeline
- Idempotent background jobs
- Transactional creation of Document + first Revision + FolderRef
- Signed internal download URLs
- Derivative file storage separated from originals
- Backup and restore strategy

The initial deployment may use self-managed object storage or a cloud object store, but the application layer must not couple document logic to one vendor.

39. API Requirements

```

GET /documents
POST /documents
GET /documents/{id}
PATCH /documents/{id}
POST /documents/{id}/revisions
GET /documents/{id}/revisions
POST /documents/{id}/folder-references
DELETE /documents/{id}/folder-references/{folderId}
POST /documents/{id}/links
DELETE /documents/{id}/links/{linkId}
POST /documents/{id}/comments
PATCH /documents/{id}/comments/{commentId}
POST /documents/{id}/approval-requests
POST /documents/{id}/promote
POST /documents/{id}/archive
POST /documents/{id}/restore
POST /documents/{id}/favorite
DELETE /documents/{id}/favorite
GET /documents/{id}/activity
GET /folders
POST /folders
PATCH /folders/{id}
POST /document-collections
POST /document-saved-views

```

Mutation endpoints must validate tenant/company scope, source-record context, revision consistency and future permission policies.

40. Frontend Architecture

- DocumentsModulePage
- FolderTree
- DocumentSummaryCards
- DocumentFilterBar
- DocumentListTable
- DocumentGrid
- DocumentTimeline
- DocumentPreviewDrawer
- DocumentPage
- DocumentHeader
- DocumentPreview
- DocumentPassportPanel
- RevisionList
- RevisionCompareView

- DocumentComments
- DocumentApprovals
- DocumentActivityTimeline
- UniversalUploadFlow
- PromoteFromTranzitFlow
- CreateShortcutFlow
- DocumentCollectionView
- StorageDashboard

40.1 Client behavior

- URL-driven folder, filters and view state
- Server-state caching
- Optimistic Star and comment updates
- Virtualized list for large folders
- Lazy loading for preview, comments and activity
- Upload queue persisted during navigation
- Keyboard shortcuts where appropriate
- Reusable design system components

41. Search and Indexing

- Index Passport metadata and linked-record names
- Index revision metadata
- Provide prefix and fuzzy matching for codes/titles
- Permission-aware result filtering later
- Near-real-time indexing after upload/revision
- Re-index job and health dashboard
- Future text extraction fields reserved in schema

42. Security and Permission-Ready Design

A detailed permission matrix is deferred, but the implementation must not hard-code universal visibility. Every document, folder, revision, comment, share and action must carry sufficient company, scope and ownership context for later enforcement.

- Tenant and company isolation
- Authenticated downloads
- File type and size validation
- Malware scan hook
- Immutable revision bytes
- Audit logging
- No approval inheritance between revisions
- No linked-record access escalation
- Private Personal Workspace
- Future confidentiality levels
- Future external participant restrictions

43. Validation and Error Handling

- Title required
- File required for first revision
- Revision code unique per document

- Folder path valid
- System folder restrictions enforced
- Duplicate checksum warning
- Unsupported preview does not block upload
- Failed preview can retry
- Upload interruption recovery
- Conflict handling for simultaneous metadata edits
- Prevent deleting a document referenced by active approvals without controlled action
- Clear per-file results in bulk upload

44. Audit and Activity Timeline

- Document created
- Metadata changed
- Folder moved
- Shortcut added/removed
- Linked record added/removed
- Revision uploaded
- Status changed
- Review requested/completed
- Approval requested/result
- Comment added/resolved
- Downloaded
- Opened
- Promoted from Transit
- Archived/restored
- Expiration changed
- Required reading assigned/acknowledged

Each activity item stores actor, timestamp, event type, readable summary and structured identifiers. Security-sensitive events should be immutable.

45. Analytics and Reporting

- Documents by company/project/module
- Documents by type/category/status
- Revision counts
- Approval turnaround
- Documents needing revision
- Expiring documents
- Required reading completion
- Storage growth
- Most opened/downloaded documents
- Transit aging
- Orphan/broken link count
- Preview failure rate

46. Performance and Scalability

- Paginated/cursor-based lists
- Virtualized folder/document rendering
- Lazy-load preview and activity
- Chunked/multipart upload for large files

- Background preview processing
- CDN or object-storage delivery abstraction
- Database indexes on company, project, folder, status, type, code and modified date
- Cache folder counts and summary cards
- Graceful degradation for unsupported preview types

Opening a document record must not wait for the full file or preview to download.

47. Testing Strategy

47.1 Unit tests

- Revision creation
- Status transitions
- Folder reference logic
- Tranzit promotion
- Duplicate checksum detection
- Archive/restore
- Expiration calculation
- Required reading state

47.2 Integration tests

- Upload to system folder
- Universal Upload with linked records
- Upload new revision
- Approval request for exact revision
- Preview worker
- Search indexing
- Activity event creation
- Orphan repair

47.3 End-to-end tests

- Create custom folder
- Upload draft to Tranzit
- Comment and mention
- Request review
- Promote to official folder
- Request approval
- Upload new revision
- Compare revisions
- Archive and restore
- Mobile upload and preview

47.4 Visual regression

- Desktop Documents Module
- Document Page
- Preview drawer
- Folder tree states
- Tablet layout
- Mobile layout
- Reduced-motion mode

48. Migration and Rollout

21. Create core tables and object-storage abstraction.
22. Migrate existing project documents into Document and Revision records.
23. Generate system folders and folder references.
24. Backfill Passport metadata from existing entities.
25. Calculate checksums and identify duplicates.
26. Generate preview jobs.
27. Validate counts, links and downloads.
28. Enable module in pilot company.
29. Monitor storage, preview failures and search indexing.
30. Roll out to all companies.

49. Acceptance Criteria

- AC-01** - Users can browse a folder tree and global document views from the same module.
- AC-02** - System and custom folders coexist.
- AC-03** - Every project automatically contains Tranzit.
- AC-04** - A Tranzit document can be promoted without creating a duplicate.
- AC-05** - One document can appear in multiple folders through shortcuts.
- AC-06** - Every document has a permanent Document Passport.
- AC-07** - Every revision is preserved and approval is revision-specific.
- AC-08** - Users can upload documents from any platform context through Universal Upload.
- AC-09** - Users can search, filter, sort, save views and Star documents.
- AC-10** - Document Page supports preview, revisions, comments, approvals and activity.
- AC-11** - Archive and restore do not destroy history.
- AC-12** - Duplicate uploads produce a clear warning.
- AC-13** - Folder and linked-record deletion does not orphan or delete documents silently.
- AC-14** - Desktop, tablet and mobile support core actions.
- AC-15** - Premium hover states, transitions and loading skeletons are implemented.
- AC-16** - Reduced-motion preferences are respected.
- AC-17** - Third-party integrations are not required for the first release.

50. Page and Flow Registry

- Documents Module Page
- Document Page
- Folder View
- Create Folder
- Universal Upload
- Upload Document
- Bulk Upload
- Upload New Revision
- Project Tranzit Folder
- Promote from Tranzit Flow
- Move Document to Official Folder
- Personal Workspace
- Document Passport

- Document Preview Drawer
- Revision History
- Revision Comparison
- Request Document Review
- Request Document Approval
- Create Document Shortcut
- Document Collection
- Required Reading
- Archive Document
- Restore Document
- Storage Dashboard
- Saved Document Views

51. Future Roadmap

- Microsoft 365 / SharePoint / OneDrive
- Google Workspace / Drive
- Email ingestion and delivery
- DocuSign / Adobe Sign / qualified signatures
- OCR and scanned-text extraction
- AI classification and folder suggestions
- Semantic AI search
- AI document summaries
- DWG/IFC semantic comparison
- Autodesk/BIM CDE integrations
- External secure sharing portal
- Records-management and legal hold enhancements

52. Visual Reference

The following approved render is the design reference for the Documents Module desktop experience. It establishes the premium visual language, layout hierarchy, warm-white surfaces, restrained color system, folder tree, status cards, document list and right-side Document Passport panel.

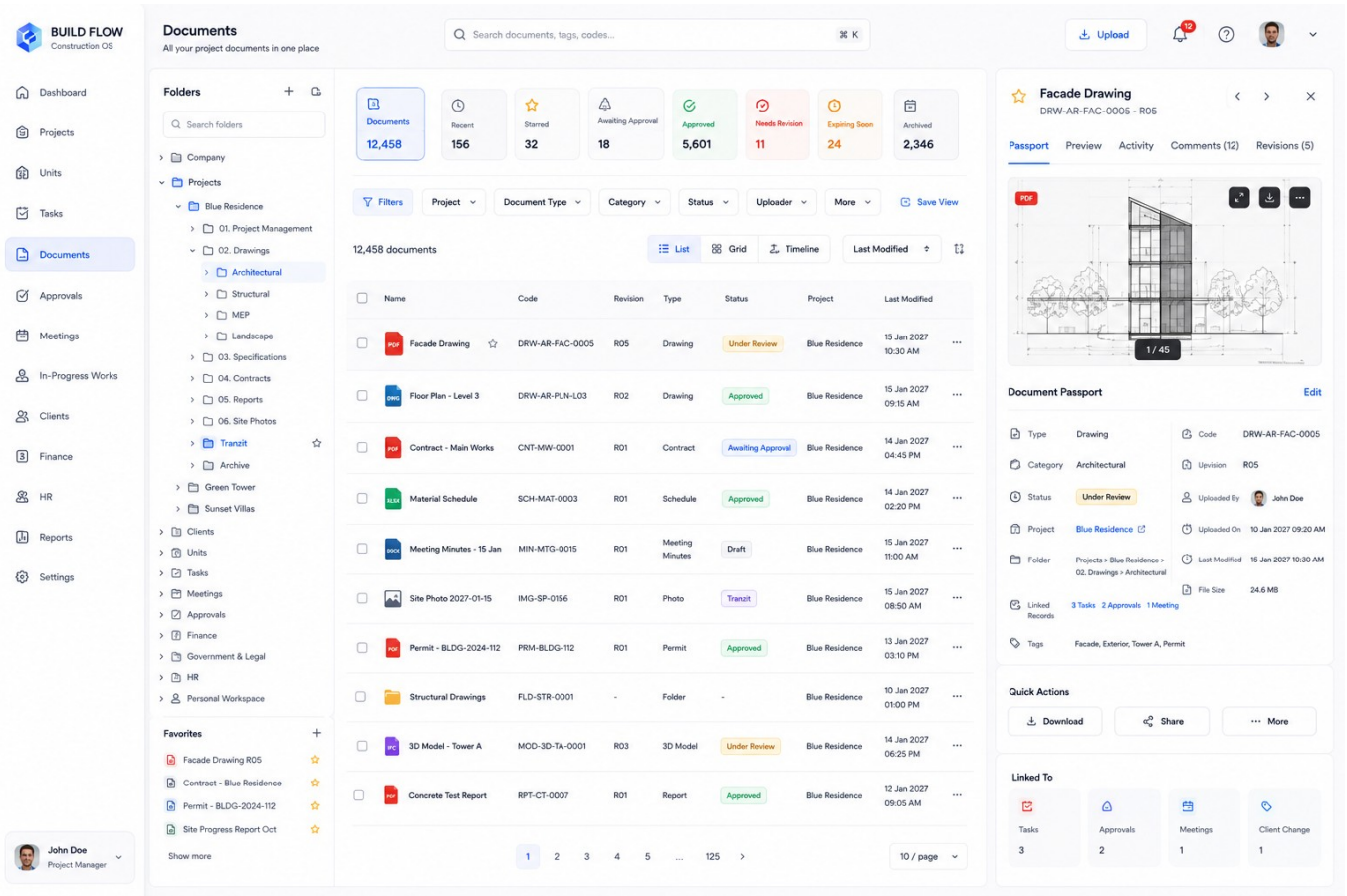


Figure 1. Confirmed premium Documents Module desktop design reference.

52.1 Visual implementation notes

- Warm white and cream surfaces with subtle borders and soft shadows.
- Gold accent reserved for primary actions, stars and selected states.
- Compact status chips with semantic colors.
- Folder tree remains visually quiet and highly scannable.
- Selected document opens in a right-side panel without losing list context.
- Rows and cards use premium hover elevation and quick-action reveal.
- Spacing and typography must remain consistent with Projects and Tasks modules.